

Developing a mobile AI system for rehabilitation to improve incorrect biomechanical movements

Problem Statement and Motivation

In physiotherapy, patients often complete exercises at home without professional supervision, which can lead to poor posture or technique that may reduce the effectiveness of therapy. This project addresses the engineering problem of monitoring at-home patient rehabilitation exercises and automatically identifying whether the performed movements are biomechanically correct or incorrect. An automated system that can classify correct versus incorrect movement provides a private, low-cost, accessible method to improve recovery outcomes.

Dataset and Solution

The dataset used in this project is **IntelliRehabDS**, an open-source rehabilitation dataset containing 3D skeleton data of patients performing physiotherapy exercises, with labels for both correct and incorrect executions. Since this dataset already contains examples of improper movement, it provides a valuable training source for machine learning models without the need for self-collected data. Since the IntelliRehab dataset consists of grayscale images data and comma-separated value data, we could apply different machine learning models to each of the datasets. For example, we could apply the convolutional neural network (CNN) to classify the rehabilitation exercises movements captured in the images into correct or incorrect. We could start with very basic neural network architecture like VGG16. However, in order to have more interpretable results, we plan to use logistic regression so that we can track which joints angles or positioning contributed more in our classification results. Eventually, we plan to apply transfer learning to fusion these two models. We will be using either Pytorch or TensorFlow to construct our machine learning models. We will also be using numpy, pandas, and matplotlib, and opencv etc. to perform data preprocessing, computational tasks, and results visualizations.

Relevance to Engineering

This solution is relevant to the field of engineering as it demonstrates how engineering and machine learning can be combined to address healthcare challenges. By leveraging sensor-based datasets and classification models, the project highlights how engineering tools can enhance remote monitoring, support patient independence, and increase the accessibility of physiotherapy care.

Appendix A – Group Contract Template

Name	Primary Contact Information
Archie Yang	archie.yang@ucalgary.ca
Glena Hussein	glena.hussein@ucalgary.ca
Danielle Blackburn	Danielle.Blackburn@ucalgary.ca
Aamena Yusuf Lwa	aamena.lwa@ucalgary.ca

1. **Communication:** What's the primary way of contacting each other remotely (text, email, etc.)?

Instagram / emails

2. **Response time:** How quickly do you expect group members to respond?

24h

3. **Meetings:** Which day and what place works best for everyone? If there is not a day or place, how do you decide when and where to meet?

Monday / Wednesday after class

4. **Division of Labour:** How will you ensure cooperation and equal/fair distribution of tasks?

Equal

5. **Accountability:** What are the team expectations regarding attendance, punctuality, participation, preparedness, task completion, deadlines, communication with the team, commitment, etc.?

notify if unavailable / any issues

6. **Decision-Making:** How will you vote on key decisions (consensus, majority, secret vote, etc.)?

7. **Conflict Resolution:** What happens when team members violate one or more terms of the contract, or their work doesn't meet the team expectations?

communicate